

ArtSoft consult

The Art of Software Development

ArtSoft Consult is a Romanian private leading digital engineering company with 140+ team members in the Central and Eastern Europe serving clients across Europe and US.

ArtSoft in a nutshell

- 100% Romanian private capital
- People centric culture
- Tier 1 private clients only
- High customer retention
- Same management team since inception

140+
Professionals

60+
Satisfied clients

80%
Revenue from Repeat Clients

300+
Completed projects

25
Years in business

History & timeline

1999–2002

Founding ArtNet

→ Established in Cluj-Napoca
→ Clients from Germany
→ Working on data mining and data qualifying projects
→ Technology stack: Java, XML, JSP, Oracle
→ Final clients: Bertelsmann, Jewish Museum from Berlin
→ Part of the R&D dept. of Caatoosee AG.
→ Managed 400 developers in Bali
→ Became Oracle Partner

2003

Rebranded to ArtSoft Consult

→ New entity ArtSoft Consult
→ Management team joined
→ New industries: automotive, telecom

2005–2006

Reached 50+ employees

→ Moved into a larger office
→ New industry: Travel
→ Technology stack: mobile apps, data and application migrations
→ Developed and implemented our own agile software development process
→ Became Microsoft partner

2008–2010

Key clients acquired

→ New key clients from Media, Digital marketing, Travel and Medical industries
→ Starting working with Scrum, Scrumban, Kanban
→ Medical software developed with artificial neural networks

2012–2013

Own office building

→ Reached 100 FTE
→ New key clients from Telecom, Education and Manufacturing industry
→ Swiss Quality Award and Forum Managed Care prizes for developed product
→Founding member of Cluj IT Cluster

2015–2016

Technology stack: Cloud and microservices

→ Reached 250 successful projects
→ Became a strategic technology partner for our clients in the Travel, Education, Medicine, Telecom and Digital marketing fields

2020–2022

150 employees

→ Reached 300 successful projects
→ Technology stack: Cloud-native, Serverless architecture, Big data
→ No specialists on the bench

2023–2024

Key clients acquired

→ Consolidation of strategic partnerships
→ ISO 27001:2022 certification
→ New key clients from Manufacturing and Commerce & Marketing industries

The same management team

Our teams are part of the R&D departments of our clients

Strategic technology partner of our clients

What we do



- Software engineering
- UX/UI and graphic design
- QA and testing
- Integration
- Security
- Roll out software products

- Data and application migrations
- Infrastructure operations
- Architecture design/redesign

- Business analysis
- Benchmarking
- Due diligence analysis
- Software architecture & design
- Optimizations
- Project management
- Security consulting

- iOS
- Android
- Cross-platform

- Digital marketing
- Customer support
- Data analysis and data management
- Business development

What we do – Domains of expertise



Enterprise applications



Portals/web applications



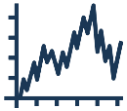
Booking engines and Loyalty systems



Telecom provisioning



Data security & Cloud



Time series



Data warehouse and business intelligence



Artificial neural networks



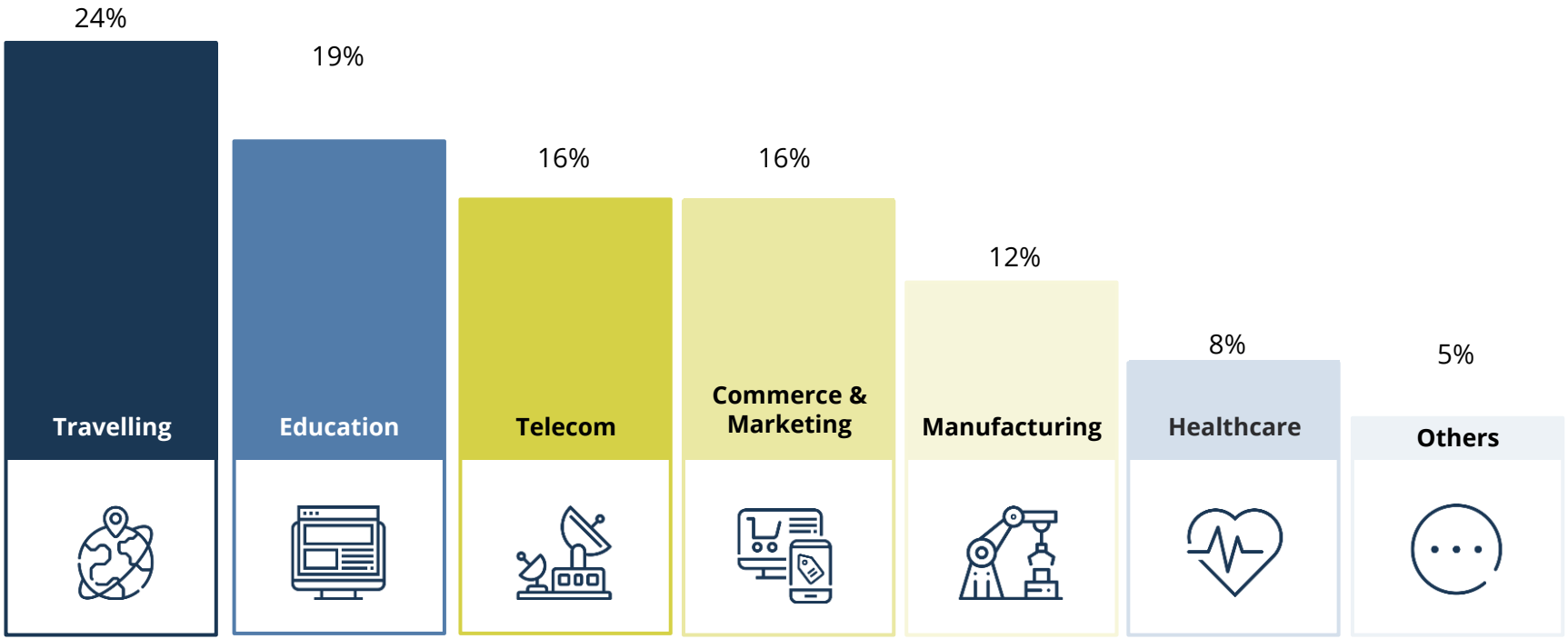
Internet of things

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Technologies (selection)



Clients by industry



Applicable to period 2020 – 2024 revenue

ETL Strategies in BI Projects

Extract

Transform

Load

ETL vs ELT

- Full Load
- Incremental (Delta) Load
- Batch ETL
- Real-Time/Streaming

Talend, Informatica, SSIS

Apache NiFi, Airbyte, dbt with timestamps

Pentaho, SSIS, AWS Glue

Apache Kafka, Spark Streaming, Fivetran (CDC)

Change Data Capture (CDC)

Essential for near real-time updates

Log based

Trigger based

Timestamp based

Method	Accuracy	Performance	Complexity	Detects Deletes
Log	High	High	Higher	Yes
Trigger	High	Medium	Medium	Yes
Timestamp	Medium	High	Low	No (only soft)

Log Based CDC

Captures changes directly from the database transaction logs.

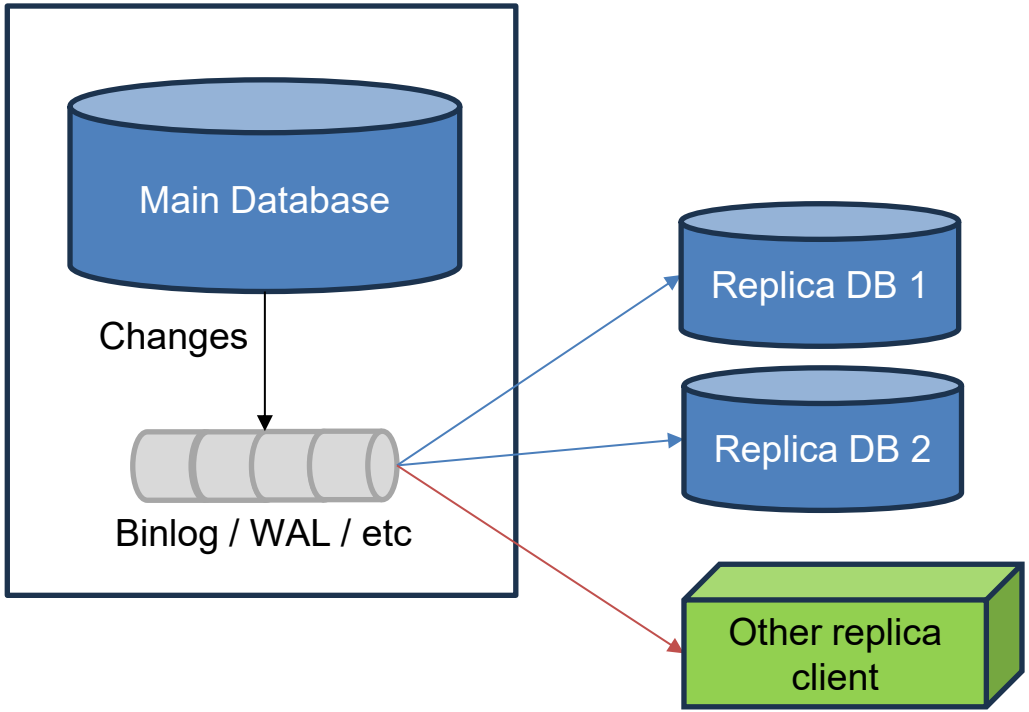
Highly efficient: it avoids querying the entire dataset.

Works well with databases supporting redo logs (MySQL binary logs, PostgreSQL WAL, SQL Server CDC, Oracle LogMiner).

Pros: Minimal performance impact, reliable, scalable.

Cons: Requires log access permissions, compatibility varies by database.

Log Based CDC



MySQL specific:

```
29  SHOW BINARY LOGS;
```

Results 1

SHOW BINARY LOGS | Enter a SQL expression to filter results (use Ctrl+Space)

	Log_name	File_size	Encrypted
1	mysql-bin.856508	111,241,998	No
2	mysql-bin.856509	110,221,102	No
3	mysql-bin.856510	104,873,480	No
4	mysql-bin.856511	105,493,156	No

```
29  SHOW BINARY LOGS;
30  show binlog events in 'mysql-bin.856508' limit 100;
```

Results 1

show binlog events in 'mysql-bin.856508' limit 100 | Enter a SQL expression to filter results (use Ctrl+Space)

		Pos	Event_type			Info
1	mysc	4	Format_desc	,294,	126	Server ver: 8.0.40-commercial, Binlog ver: 4
2	mysc	126	Previous_gtids	,294,	157	
3	mysc	157	Anonymous_Gtid	,294,	237	SET @@SESSION.GTID_NEXT= 'ANONYMOUS'
4	mysc	237	Query	,294,	333	BEGIN
5	mysc	333	Table_map	,294,	469	table_id: 276 (availability_engine.ae_activity)
6	mysc	469	Write_rows	,294,	8,616	table_id: 276

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Debezium: CDC for Real-Time Data Streaming



Open-source distributed CDC platform.

Captures changes in databases without modifying application logic.

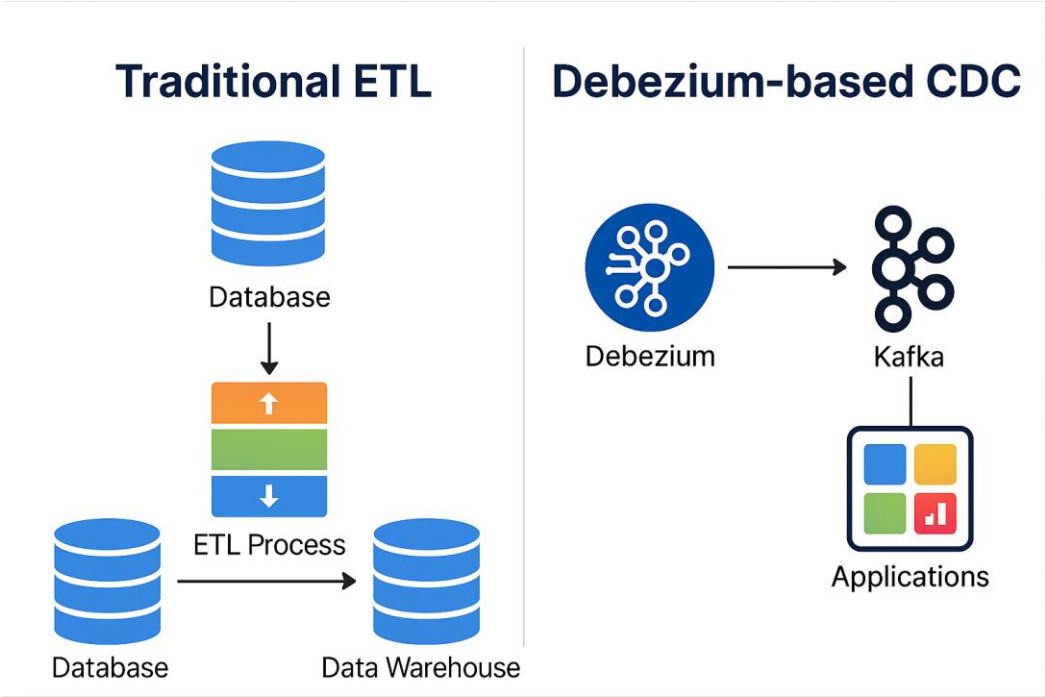
Ideal for event-driven architectures.

Enables real-time ETL and analytics.

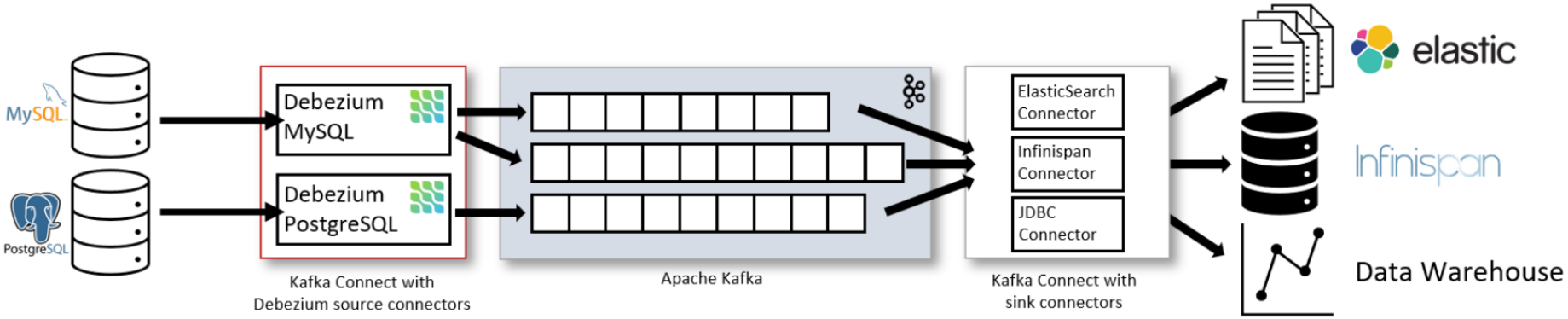
Reduces database polling inefficiencies.

Debezium Core Components

- Source Connectors (database-specific capture agents)
- Kafka Topics (CDC event streams)
- Consumer Applications (BI tools & analytics platforms)



Debezium Architecture



Debezium: Incremental Snapshots

By default gets a full snapshot of the tables. When having huge tables, this snapshot process can take a lot of time and in the meantime the data change events are suppressed. A more flexible approach is to use incremental snapshots. For this to work, there are several prerequisites:

- **primary key** on source table
- **snapshot.mode: no_data** - prevent the initial snapshot
- **signal.data.collection: <signal_table_name>** - the “signal table”

table.include.list: ……,<signal_table_name>,…. - add the “signal table” to the list of tables that will be monitored for changes;

- **incremental.snapshot.watermarking.strategy** - **insert_delete**

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Debezium Advantages

1. Transactional Event Ordering: events are streamed in the order of db transactions.
2. Schema Evolution Tracking: detects changes in database schemas.
3. Exactly-Once Message Semantics: avoid duplicate messages.
4. Embedded Mode: can run inside an application instead of using Kafka Connect.
5. Filtering & Data Masking: fine-grained filtering to capture only relevant changes.
6. Automatic Recovery & Reconciliation: detects inconsistencies and automatically resynchronizes missing or failed events.
7. Snapshotting Mechanism: incremental snapshots to backfill historical records.
8. Low-Latency Streaming : works with Kafka Streams, Flink, and Apache Beam for low-latency analytics.

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